Regression and Model Fitting

LumoraAI Presentation Demo • 04 Machine Learning

Regression

• Regression predicts a continuous target from input features. Linear regression models the target as a weighted combination of features plus an intercept.

Loss and optimization

• Mean squared error averages squared prediction errors. Gradient-based optimization adjusts parameters in the direction that reduces the loss, with the learning rate controlling step size.

Generalization

• A model that fits training data extremely well can still perform poorly on unseen examples. Regularization, validation data and appropriate model complexity can help control overfitting.

Demo resource generated specifically for the LumoraAI presentation environment. The content is original and intended for RAG retrieval, quiz generation, and citation demos.